MATER ACADEMY CHARTER HIGH SCHOOL

Biology 1 End-of-Course Simulation
Assessment: Progress Monitoring 2 (PM2)

School Year: 2025–2026 | Course: Biology 1

Total Questions: 60 Multiple Choice | Time Allowed: 160 Minutes

Format: Paper-Based Accommodation Version (IEP/504)

A2 Systems Architecture | Confidential — Not for distribution outside Mater Academy

STUDENT INFORMATION

Student Name: _____ Date: _____

Student ID: _____ Period: _____

Teacher: _____

INSTRUCTIONS

1. This assessment contains 60 multiple-choice questions. Each question has four answer choices (A, B, C, D).
2. Select the ONE best answer. Circle or clearly mark your answer on this paper.
3. Do NOT use a calculator unless your IEP or 504 Plan specifically permits one.
4. A Periodic Table of the Elements is provided separately if needed.
5. Work through all 60 questions. If unsure, make your best choice and continue.
6. Do not discuss questions with other students during the assessment.
7. When finished, return this paper to your teacher. Do NOT remove it from the room.

BLUEPRINT

Reporting Category	Content Area	Questions	Weight
RC1	Molecular and Cellular Biology	21	35%
RC2	Classification, Heredity, and Evolution	15	25%
RC3	Organisms, Populations, and Ecosystems	24	40%

RC1 — Molecular and Cellular Biology | 21 items | 35%

Question 1 RC1 — Molecular and Cellular Biology

A student observes a thin slice of cork under a microscope and sees small, box-like compartments. This observation directly supports which component of the cell theory?

A. All cells require oxygen to survive.
B. Cells produce energy through photosynthesis.
C. All cells contain a nucleus.
D. All living things are composed of cells.

Question 2 RC1 — Molecular and Cellular Biology

A scientist proposes that life can arise spontaneously from nonliving matter. Which of the following findings would most directly challenge this claim according to the modern cell theory?

A. Cells contain DNA that controls their functions.
B. Most cells are too small to be seen with the naked eye.
C. Cells vary in size and shape depending on their function.
D. All observed cells have originated from pre-existing cells.

Question 3 RC1 — Molecular and Cellular Biology

A researcher designs an experiment to test whether a new antibiotic kills bacteria by disrupting their cell membranes. She exposes two groups of bacteria to the antibiotic — one group in a hypertonic solution and one in a hypotonic solution — and measures cell survival. Which of the following best describes why this experimental design is valid?

A. It tests only one independent variable while keeping all other conditions constant.
B. It includes a variable that affects the dependent variable being measured.
C. It eliminates all possible sources of error in the investigation.
D. It uses the same bacterial species in both groups, controlling for organism type.

Question 4 RC1 — Molecular and Cellular Biology

Which of the following structures is found in prokaryotic cells but NOT in eukaryotic cells?

A. Cell membrane
B. Plasmid
C. Ribosome
D. Cytoplasm

INTER ACADEMY CHARTER HIGH SCHOOL
Inter Academy Charter High School

Question 5 RC1 — Molecular and Cellular Biology

A plant cell placed in a solution begins to lose water and the cell membrane pulls away from the cell wall. Which of the following best explains this observation?

A. The solution outside the cell has a lower solute concentration than the cell's interior.
B. The cell wall is actively pumping water out of the cell.
C. The cell membrane has become impermeable to water molecules.
D. The solution outside the cell has a higher solute concentration than the cell's interior.

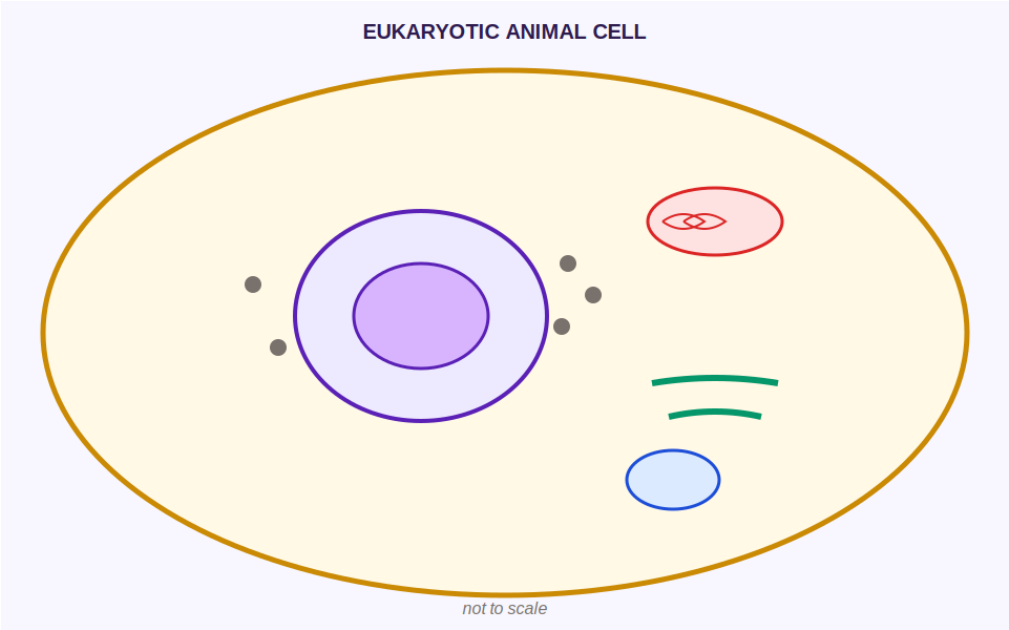
Question 6 RC1 — Molecular and Cellular Biology

A liver cell and a muscle cell in the same organism contain the same DNA, yet they perform very different functions. Which of the following best explains this difference?

A. Different cells contain different numbers of chromosomes.
B. Mutations in the DNA of each cell type cause different proteins to form.
C. Different genes are expressed in different cell types.
D. Each cell type receives different nutrients from the bloodstream.

Question 7 RC1 — Molecular and Cellular Biology

A cell biologist observes that a cell has a large number of mitochondria, an extensive Golgi apparatus, and numerous secretory vesicles. Based on these observations, which of the following conclusions about this cell is most strongly supported?



A. The cell is highly active in producing and secreting proteins.
B. The cell is capable of performing photosynthesis.
C. The cell primarily stores energy in the form of starch granules.
D. The cell undergoes rapid cell division to replace damaged tissue.

Question 8 RC1 — Molecular and Cellular Biology

Which of the following processes moves a substance across the cell membrane WITHOUT requiring the cell to use energy?

A. Endocytosis
B. Facilitated diffusion
C. Active transport
D. Sodium-potassium pump

Question 9 RC1 — Molecular and Cellular Biology

Red blood cells placed in a solution that is less concentrated than their interior begin to swell. Which of the following best predicts what will happen to these cells over time if left in this solution?

A. The cells will shrink as water moves out through the membrane.
B. The cells will remain unchanged because the membrane blocks water movement.
C. The cells may burst as water continues to move into the cells by osmosis.
D. The cells will actively pump solutes out to restore equilibrium.

Question 10 RC1 — Molecular and Cellular Biology

Which of the following correctly matches a macromolecule to its primary function in living organisms?

A. Lipids — storing and transmitting genetic information
B. Proteins — providing long-term energy storage in seeds
C. Carbohydrates — providing a rapid source of energy for cells
D. Nucleic acids — catalyzing biochemical reactions

Question 11 RC1 — Molecular and Cellular Biology

An enzyme that normally breaks down starch stops functioning when placed in a highly acidic solution. Which of the following best explains this observation?

A. The acid breaks apart the starch molecules before the enzyme can act.
B. The change in pH alters the shape of the enzyme's active site.
C. The enzyme requires a basic environment to bind its substrate.
D. The acidic solution causes the starch to become a different type of molecule.

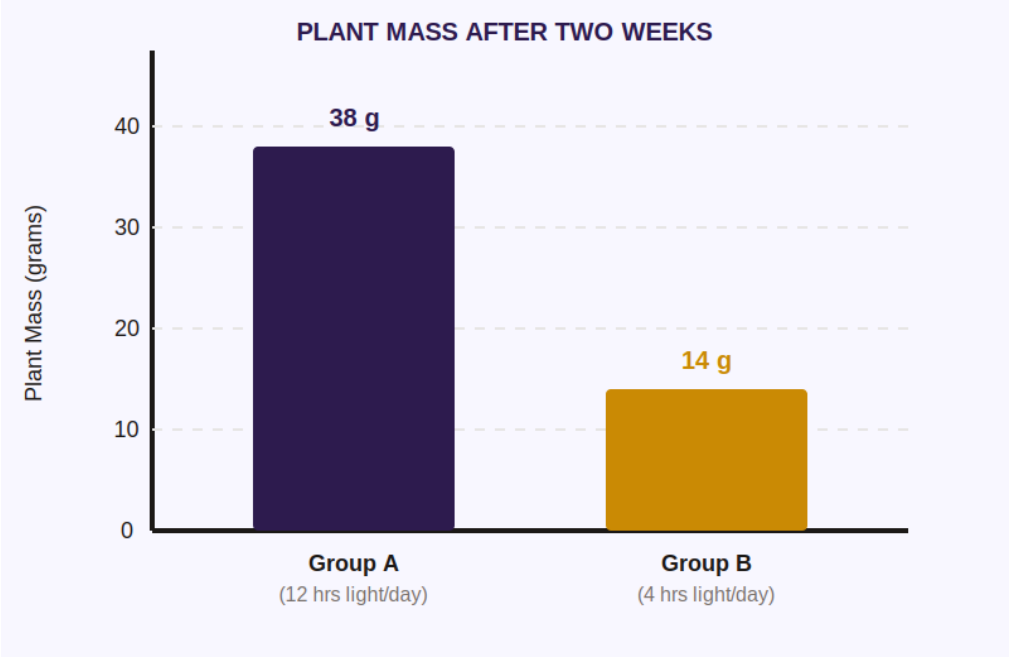
Question 12 RC1 — Molecular and Cellular Biology

During photosynthesis, plants use light energy to convert carbon dioxide and water into glucose and oxygen. During cellular respiration, the reverse process occurs. Which of the following correctly describes the relationship between these two processes?

A. Photosynthesis produces ATP directly for use during cellular respiration.
B. Cellular respiration releases the energy stored in glucose during photosynthesis.
C. Both processes occur simultaneously in the chloroplast of plant cells.
D. Photosynthesis and cellular respiration both require oxygen as a reactant.

Question 13 RC1 — Molecular and Cellular Biology

A scientist grows two groups of aquatic plants under identical conditions except for light exposure. Group A receives 12 hours of light per day, and Group B receives 4 hours. After two weeks, Group A plants have significantly greater mass. Which of the following best explains this difference?



A. Group A plants absorbed more minerals from the water due to greater root activity.
B. Group B plants used more energy for cellular respiration, slowing their growth.
C. Group A plants produced more glucose through photosynthesis, supporting greater growth.
D. Group A plants stored more water in their cells because of increased light exposure.

Question 14 RC1 — Molecular and Cellular Biology

During DNA replication, a strand with the base sequence ATCGGA is used as a template. What is the complementary base sequence of the new strand produced?

A. ATCGGA
B. TAGCCT
C. UAGCCU
D. TACGGA

Question 15 RC1 — Molecular and Cellular Biology

A mutation causes a single base substitution in a gene. The resulting mRNA codon still codes for the same amino acid as the original codon. Which of the following best explains why this mutation does NOT change the protein produced?

A. The mutation occurs in a non-coding region of the DNA.
B. The ribosome repairs the mutated codon during translation.
C. The genetic code is degenerate, meaning multiple codons can code for the same amino acid.
D. DNA replication corrects single-base errors before transcription occurs.

Question 16 RC1 — Molecular and Cellular Biology

A segment of DNA has the template strand sequence: 3'-TACGCAATT-5'. What is the sequence of the mRNA produced during transcription from this template?

A. 3'-AUGCGUUAA-5'
B. 5'-TACGCAATT-3'
C. 5'-AUGCGUUAA-3'
D. 3'-UACGCAAUU-5'

Question 17 RC1 — Molecular and Cellular Biology

A skin cell in a human contains 46 chromosomes. After undergoing mitosis, how many chromosomes will each daughter cell contain, and what type of reproduction does this represent?

A. 23 chromosomes; sexual reproduction
B. 92 chromosomes; sexual reproduction
C. 23 chromosomes; asexual reproduction
D. 46 chromosomes; asexual reproduction

Question 18 RC1 — Molecular and Cellular Biology

A researcher discovers that a chemical agent prevents homologous chromosomes from separating during meiosis I. Which of the following would be the most likely consequence in the resulting gametes?

A. The gametes would contain the correct number of chromosomes but with altered gene sequences.
B. Fertilization would be prevented because the gametes could not fuse with other gametes.
C. The gametes would undergo an additional round of division to restore normal chromosome number.
D. The gametes would have either too many or too few chromosomes, leading to abnormal offspring.

Question 19 RC1 — Molecular and Cellular Biology

A chromosomal mutation results in a segment of chromosome 5 being transferred to chromosome 12. This type of mutation is best described as which of the following?

A. A point mutation affecting a single nucleotide
B. A translocation affecting chromosome structure
C. A frameshift mutation caused by a deletion
D. An inversion that reverses a chromosomal segment

Question 20 RC1 — Molecular and Cellular Biology

Cancer cells divide uncontrollably and do not respond to normal regulatory signals. A drug is developed that reactivates proteins that normally signal a cell to stop dividing. Which of the following best explains how this drug would affect tumor growth?

A. The drug would slow or stop tumor growth by restoring normal cell cycle regulation.
B. The drug would cause cancer cells to undergo apoptosis by activating cell death pathways.
C. The drug would reverse mutations in the cancer cell's DNA, returning it to a normal state.
D. The drug would prevent metastasis by blocking cancer cells from entering the bloodstream.

Question 21 RC1 — Molecular and Cellular Biology

Water molecules are able to form hydrogen bonds with each other. Which of the following properties of water is MOST directly explained by hydrogen bonding between water molecules?

A. Water has a lower density as a solid than as a liquid.
B. Water is an effective solvent for ionic compounds.
C. Water has a high specific heat capacity.
D. Water molecules are polar.

RC2 — Classification, Heredity, and Evolution | 15 items | 25%

Question 22 RC2 — Classification, Heredity, and Evolution

A newly discovered organism is unicellular, has no membrane-bound nucleus, and can survive in extremely hot, acidic environments. Into which domain should this organism most likely be classified?

A. Archaea
B. Eukarya
C. Bacteria
D. Protista

Question 23 RC2 — Classification, Heredity, and Evolution

Scientists recently reclassified a species of organism from one kingdom to another based on new genetic evidence. Which of the following best explains why scientists changed this organism's classification?

A. The organism developed new physical characteristics that matched a different kingdom.
B. New evidence revealed that the organism's evolutionary relationships were different than previously thought.
C. The original classification system was replaced with a completely new system.
D. Scientists disagreed about which physical traits should be used to classify the organism.

Question 24 RC2 — Classification, Heredity, and Evolution

In pea plants, the allele for tall stems (T) is dominant over the allele for short stems (t). A homozygous tall plant is crossed with a homozygous short plant. What proportion of the F1 offspring will be tall?

A. 0% (none)
B. 100%
C. 25%
D. 50%

Question 25 RC2 — Classification, Heredity, and Evolution

In humans, the gene for color blindness is recessive and located on the X chromosome. A woman who is a carrier for color blindness has children with a man who has normal color vision. Which of the following correctly describes the expected outcome for their sons?

A. All sons will have normal color vision.
B. All sons will be color blind.
C. 50% of sons will be carriers for color blindness.
D. 50% of sons will be color blind.

Question 26 RC2 — Classification, Heredity, and Evolution

In a population of cattle, neither the red coat allele nor the white coat allele is dominant. Cattle heterozygous for coat color display a roan coat — a mix of red and white hairs. A red bull is crossed with a white cow. What is the expected ratio of coat colors in the offspring?

A. All offspring will be roan.
B. 1 red : 1 white
C. 3 red : 1 white
D. 1 red : 2 roan : 1 white

Question 27 RC2 — Classification, Heredity, and Evolution

A couple has three children, all of whom have attached earlobes. Both parents have free earlobes. A fourth child is born with attached earlobes. The allele for attached earlobes is recessive. Which of the following best explains the inheritance pattern observed in this family?

A. Both parents are homozygous dominant for the earlobe trait.
B. Both parents are heterozygous carriers of the recessive allele.
C. The attached earlobe trait is the result of a new mutation in each child.
D. Attached earlobes are dominant because they appear in all four children.

Question 28 RC2 — Classification, Heredity, and Evolution

The Miller-Urey experiment simulated conditions thought to exist on early Earth and produced organic molecules from inorganic compounds. Which of the following conclusions is most directly supported by this experiment?

A. Life on Earth originated from organisms that arrived on meteorites.
B. The first living cells formed in hydrothermal vents deep in the ocean.
C. RNA was the first molecule capable of storing genetic information on early Earth.
D. The chemical building blocks of life could have formed naturally from nonliving matter.

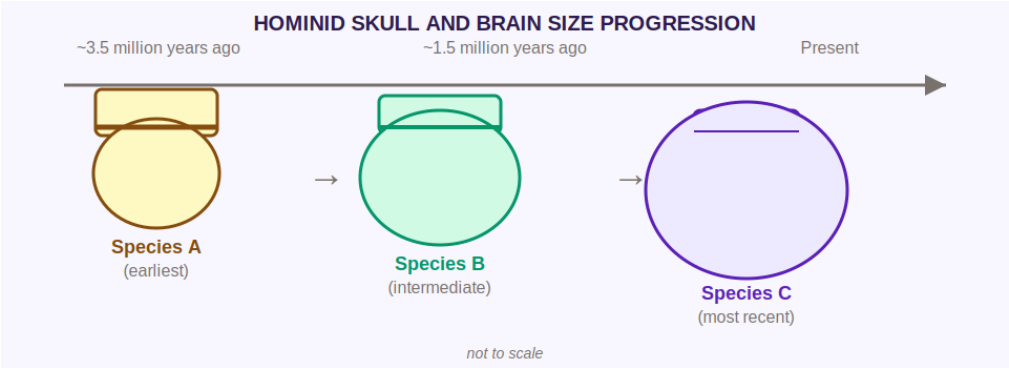
Question 29 RC2 — Classification, Heredity, and Evolution

The forelimbs of whales, bats, and humans contain the same basic bone structures arranged in similar patterns, despite being used for very different functions. Which of the following best explains this observation as evidence for evolution?

A. These organisms independently evolved similar structures to perform similar functions.
B. Environmental pressures caused these organisms to develop identical forelimb bones.
C. These organisms share a common ancestor from which they inherited similar bone structures.
D. These organisms exchanged genetic material during their evolutionary history.

Question 30 RC2 — Classification, Heredity, and Evolution

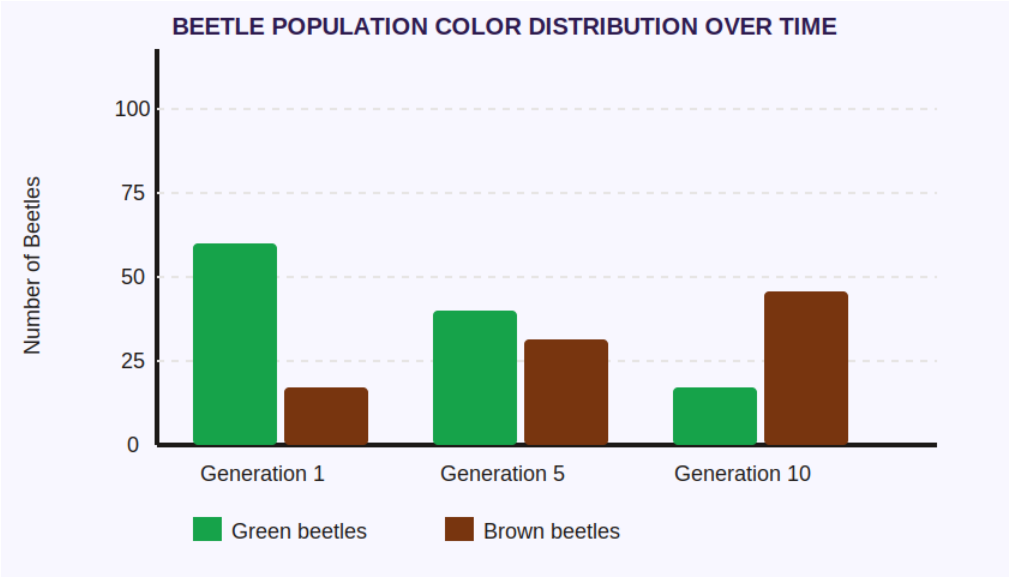
Fossil records show that the average brain size of hominids increased significantly over millions of years. Which of the following trends in hominid evolution is most consistent with this observation?



A. Hominids gradually developed the ability to walk on two legs.
B. Hominids developed increasingly complex tools and language over time.
C. Hominids evolved smaller jaw sizes to accommodate a more varied diet.
D. Hominids migrated from Africa to other continents over time.

Question 31 RC2 — Classification, Heredity, and Evolution

In a population of beetles, green individuals are more easily spotted by predators than brown individuals on a forest floor. Over several generations, the proportion of brown beetles increases significantly. Which of the following conditions of natural selection does this scenario best illustrate?



A. Overproduction of offspring leads to competition for limited resources.
B. Genetic drift causes random changes in allele frequencies in the population.
C. Heritable variation in color affects the survival and reproduction of individuals.
D. Gene flow introduces new color alleles from a neighboring beetle population.

Question 32 RC2 — Classification, Heredity, and Evolution

A small group of birds is blown by a storm to an isolated island. The founding population happens to carry a rare allele at high frequency purely by chance. Over many generations, this allele becomes common throughout the island population even though it provides no survival advantage. Which mechanism of evolutionary change best explains this outcome?

A. Natural selection favoring the rare allele in the island environment
B. Genetic drift causing random changes in allele frequency in the small population
C. Gene flow introducing the allele repeatedly from the mainland population
D. Mutation increasing the rate at which the allele is produced in offspring

Question 33 RC2 — Classification, Heredity, and Evolution

A bacterial population is exposed to a low dose of antibiotic. Most bacteria die, but a small number with a resistance mutation survive and reproduce. After several generations without antibiotics, the population recovers. When exposed to the same antibiotic again, nearly all bacteria survive. Which of the following best explains this outcome?

A. The bacteria learned to tolerate the antibiotic through repeated exposure.
B. The antibiotic caused new resistance mutations to form in the surviving bacteria.
C. Genetic recombination during reproduction produced resistance in all offspring.
D. Natural selection increased the frequency of the resistance allele in the population.

Question 34 RC2 — Classification, Heredity, and Evolution

Scientists inserted the human gene for insulin production into bacteria, which then began producing human insulin. Which of the following best explains why the bacteria were able to use the human gene to produce functional insulin?

A. The genetic code is nearly universal, so bacteria can translate human genes into protein.
B. Bacteria and humans share the same ribosomes, which can translate any gene.
C. Human genes are simpler than bacterial genes, making them easier to express.
D. Bacteria naturally produce insulin-like molecules that are modified by the human gene.

Question 35 RC2 — Classification, Heredity, and Evolution

Artificial selection in dogs has produced breeds with dramatically different sizes, coat types, and temperaments over a relatively short period of time. Which of the following best explains how artificial selection differs from natural selection?

A. Artificial selection requires genetic mutations to occur at a faster rate than natural selection.
B. In artificial selection, humans choose which traits are favored rather than the environment.
C. Artificial selection can only produce changes in physical traits, not behavioral ones.
D. Natural selection acts on individuals, while artificial selection acts on entire populations.

Question 36 RC2 — Classification, Heredity, and Evolution

A researcher uses a technique to produce genetically identical copies of a sheep by inserting the nucleus of an adult sheep's cell into an egg cell that had its nucleus removed. The resulting organism is a clone. Which of the following best describes the genetic relationship between the clone and the original sheep?

A. The clone has half the genetic information of the original sheep.
B. The clone is genetically identical to both the egg donor and the original sheep.
C. The clone has all new genetic information produced by the cloning process.
D. The clone has identical nuclear DNA but may have different mitochondrial DNA.

RC3 — Organisms, Populations, and Ecosystems | 24 items | 40%

Question 37 RC3 — Organisms, Populations, and Ecosystems

Guard cells surround the stomata on the surface of a leaf. Which of the following best describes the role of stomata in plant physiology?

A. They absorb water from the soil and transport it upward through the stem.
B. They produce glucose that is transported to other parts of the plant.
C. They protect the leaf from damage by closing during periods of high light intensity.
D. They allow for the exchange of gases needed for photosynthesis and cellular respiration.

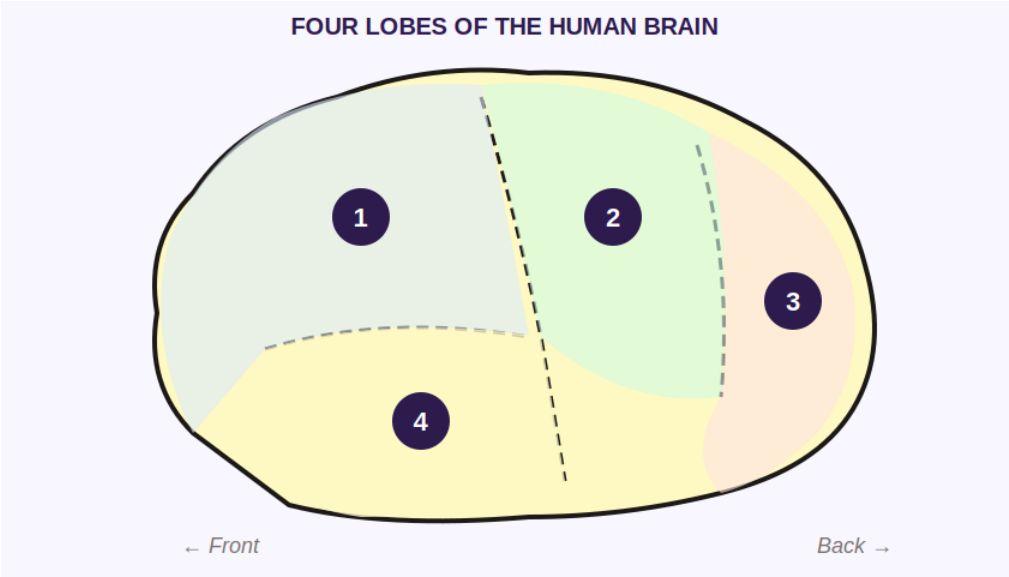
Question 38 RC3 — Organisms, Populations, and Ecosystems

A plant in a dry environment has very few stomata on its leaf surfaces and a thick, waxy cuticle coating. Which of the following best explains how these structural adaptations benefit the plant?

A. They increase the rate of photosynthesis by trapping more carbon dioxide inside the leaf.
B. They allow the plant to absorb water directly from the air through its leaf surface.
C. They protect the plant from herbivores by making leaves difficult to chew.
D. They reduce water loss through transpiration, helping the plant survive in arid conditions.

Question 39 RC3 — Organisms, Populations, and Ecosystems

The diagram below shows four lobes of the human brain labeled 1 through 4, from front to back. Which lobe, located at the front of the brain, is primarily associated with decision-making and voluntary movement?



A. Label 1 — Frontal lobe
B. Label 2 — Parietal lobe
C. Label 3 — Temporal lobe
D. Label 4 — Occipital lobe

Question 40 RC3 — Organisms, Populations, and Ecosystems

During vigorous exercise, blood flow to the skeletal muscles increases significantly. Which of the following best explains the cardiovascular mechanism responsible for this change?

A. Blood vessels in the muscles dilate, reducing resistance and increasing blood flow.
B. Blood pressure decreases, allowing more blood to flow passively to the muscles.
C. The heart rate slows to allow more blood to fill the chambers between beats.
D. Red blood cell production increases rapidly to carry more oxygen to muscles.

Question 41 RC3 — Organisms, Populations, and Ecosystems

A person is exposed to a pathogen for the first time and develops an immune response. When exposed to the same pathogen years later, the immune response is much faster and stronger. Which of the following best explains this observation?

A. The immune system produces more nonspecific defenses after the initial infection.
B. The pathogen is weaker during the second exposure due to natural changes in its structure.
C. Memory B cells produced during the first exposure respond rapidly upon re-exposure.
D. Interferons produced during the first infection remain active in the body for years.

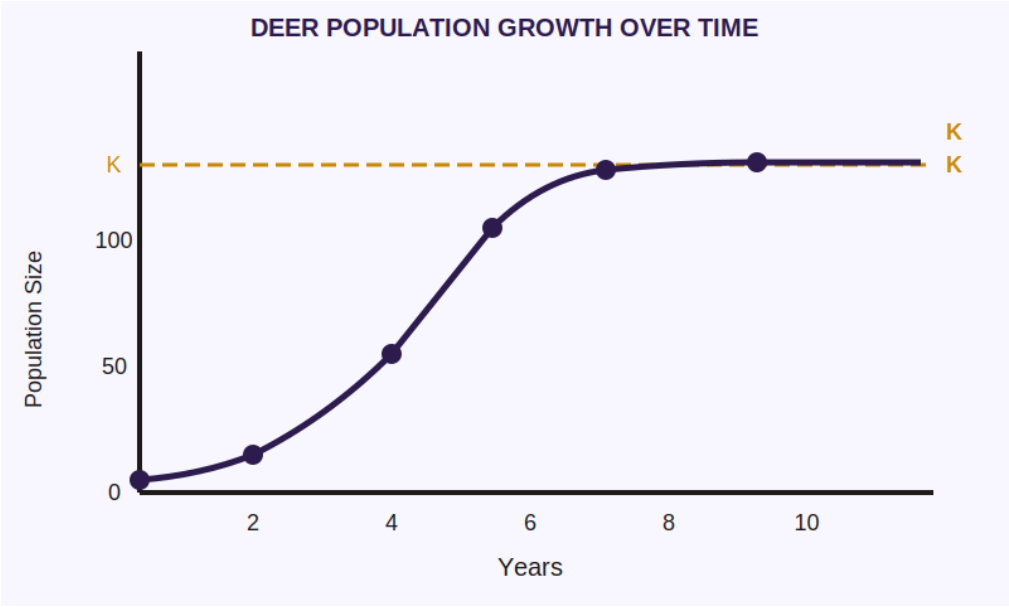
Question 42 RC3 — Organisms, Populations, and Ecosystems

A patient has a genetic condition in which their immune system produces antibodies that attack the body's own joint tissue. A doctor recommends a treatment that suppresses the immune system. Which of the following best describes a potential risk of this treatment?

A. The patient may develop allergies to substances that were previously harmless.
B. The patient may become more susceptible to infections because immune defenses are reduced.
C. The patient's immune system may begin attacking red blood cells instead of joint tissue.
D. The patient may experience uncontrolled cell growth because immune surveillance is reduced.

Question 43 RC3 — Organisms, Populations, and Ecosystems

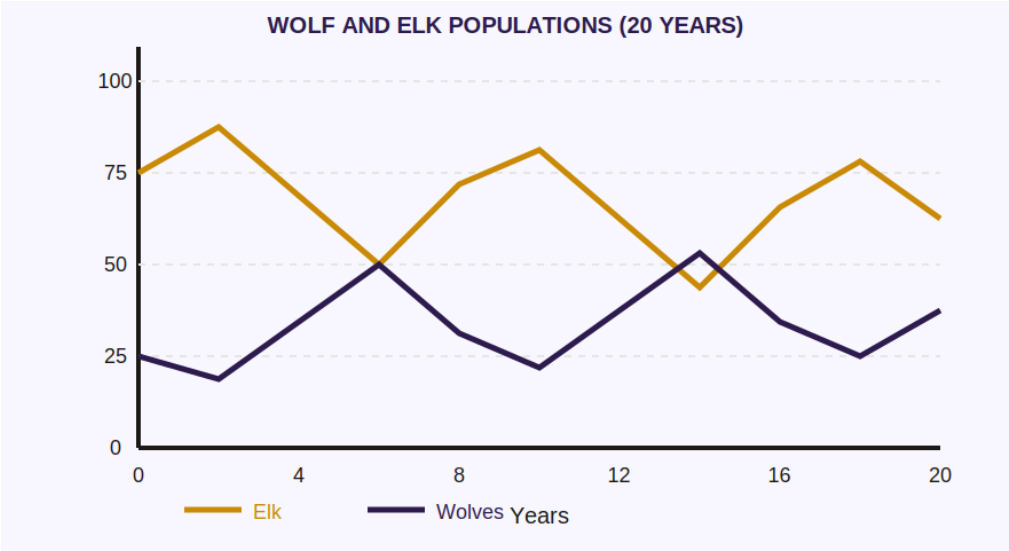
A population of deer in a forest grows rapidly until it reaches a point where food becomes scarce and disease increases. The population then stabilizes at a lower level. Which of the following concepts best describes the population level at which growth stabilizes?



A. Carrying capacity
B. Exponential growth rate
C. Biotic potential
D. Population density

Question 44 RC3 — Organisms, Populations, and Ecosystems

The graph below shows the population sizes of wolves and elk in a national park over 20 years. When wolf populations increase, elk populations decrease, and vice versa. Which of the following ecological relationships best explains this pattern?



A. Predation — wolves consume elk, limiting elk population growth.
B. Mutualism — both species benefit from the interaction.
C. Commensalism — one species benefits while the other is unaffected.
D. Competition — wolves and elk compete for the same food resources.

Question 45 RC3 — Organisms, Populations, and Ecosystems

A national park introduces wolves as a predator to control the overpopulation of elk. Over the next decade, vegetation along riverbanks increases, beaver populations grow, and river flow patterns change. Which of the following best explains this chain of ecological effects?

A. Wolves directly consumed the beaver population, reducing competition with elk for riverbank plants.
B. The introduction of wolves caused elk to migrate, leaving the park and allowing vegetation to grow.
C. Wolves reduced elk grazing pressure on riverbank vegetation, allowing plant recovery that supported beavers.
D. Wolves altered the river flow by digging dens near riverbanks, creating new habitat for beavers.

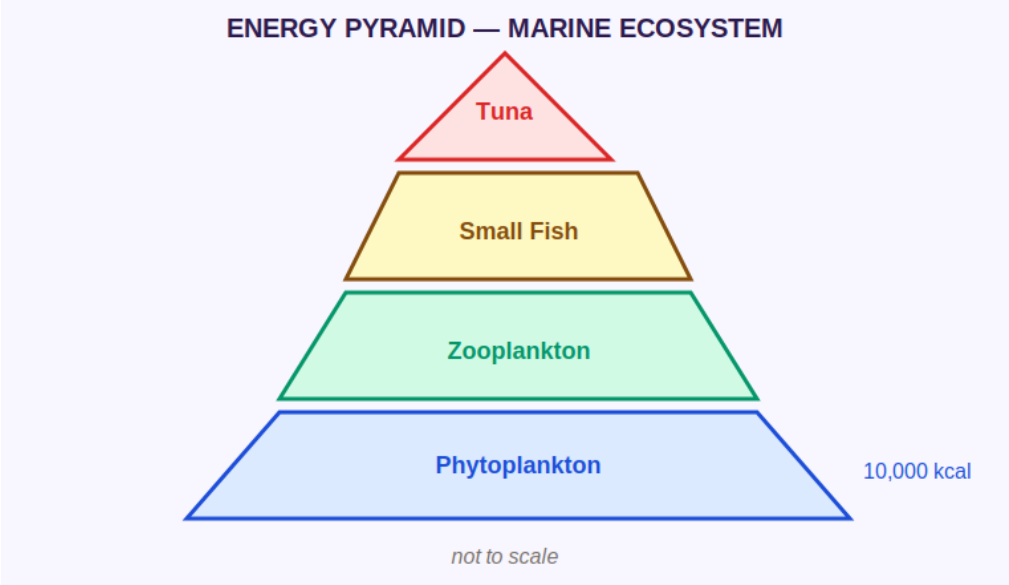
Question 46 RC3 — Organisms, Populations, and Ecosystems

In a grassland ecosystem, grass is eaten by grasshoppers, which are eaten by frogs, which are eaten by snakes. Which of the following organisms occupies the highest trophic level in this food chain?

A. Snake
B. Grass
C. Grasshopper
D. Frog

Question 47 RC3 — Organisms, Populations, and Ecosystems

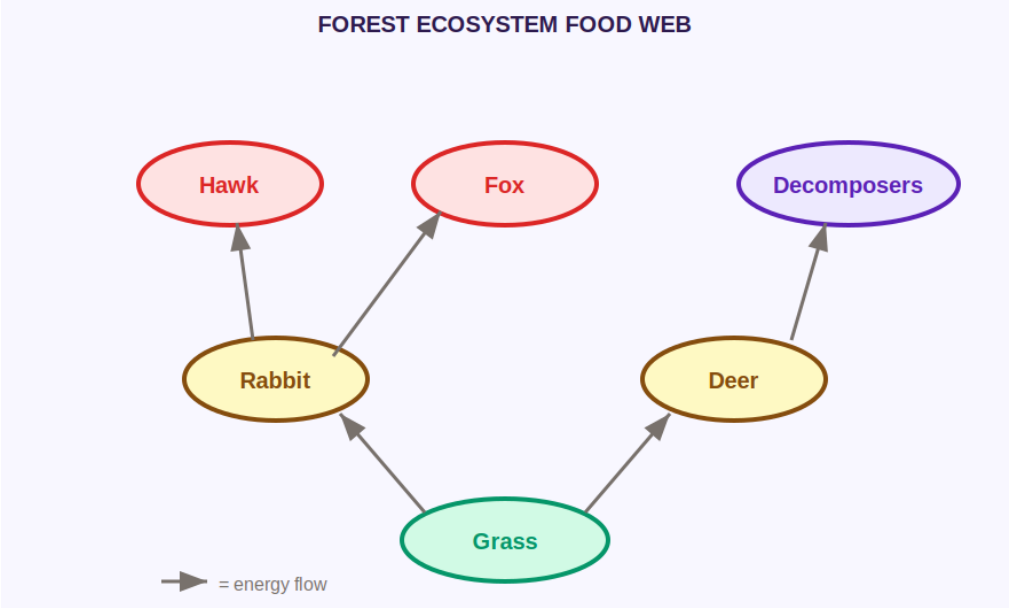
In a marine ecosystem, only about 10% of the energy available at one trophic level is transferred to the next. A food chain contains phytoplankton → zooplankton → small fish → tuna. If the phytoplankton level contains 10,000 kilocalories of energy, approximately how much energy is available to tuna?



A. 1,000 kilocalories
B. 100 kilocalories
C. 1 kilocalorie
D. 10 kilocalories

Question 48 RC3 — Organisms, Populations, and Ecosystems

A disease kills 90% of the rabbit population in an ecosystem. Rabbits are the primary prey of hawks and foxes, and also compete with deer for grass. Which of the following best predicts the long-term effects of this disease on the ecosystem?



A. Hawk and fox populations will increase because they will switch to eating deer.
B. Deer populations will decrease due to increased competition with the remaining rabbits.
C. Grass will initially increase as rabbit grazing pressure is reduced, and hawk and fox populations will decline.
D. The ecosystem will remain stable because deer will fill the ecological role of rabbits.

Question 49 RC3 — Organisms, Populations, and Ecosystems

Carbon is continuously cycled through an ecosystem. Which of the following correctly describes how carbon moves from the atmosphere into living organisms?

A. Producers absorb carbon dioxide from the atmosphere during photosynthesis and convert it to organic compounds.
B. Decomposers break down organic matter and release carbon into the soil, where plants absorb it.
C. Consumers absorb carbon dioxide directly through their respiratory surfaces.
D. Carbon enters living organisms when they drink water that contains dissolved carbon dioxide.

Question 50 RC3 — Organisms, Populations, and Ecosystems

Nitrogen is an essential element for building proteins and nucleic acids. Most organisms cannot use nitrogen gas (N_2) directly from the atmosphere. Which of the following correctly describes how nitrogen becomes available to plants?

A. Nitrogen-fixing bacteria in the soil convert nitrogen gas into ammonia that plants can absorb.
B. Plants absorb nitrogen gas through their stomata and convert it during photosynthesis.
C. Decomposers release nitrogen gas from organic matter directly into plant roots.
D. Lightning strikes convert nitrogen gas into glucose that dissolves in rainwater.

INTERNATIONAL CHARTER HIGH SCHOOL
Internal Use Only

Question 51 RC3 — Organisms, Populations, and Ecosystems

Deforestation of a tropical rainforest leads to the extinction of several species of insects, birds, and plants. Which of the following best explains why the loss of these species affects the stability of the remaining ecosystem?

A. The surviving species will evolve rapidly to fill the ecological roles of the extinct species.
B. Extinction of species in one area has no effect on ecosystem stability because other species compensate.
C. The loss of large predators destabilizes ecosystems more than the loss of insects or plants.
D. The loss of species reduces the number of ecological interactions, making the ecosystem less resilient to further disturbances.

Question 52 RC3 — Organisms, Populations, and Ecosystems

Atmospheric carbon dioxide levels have increased significantly over the past 150 years due to fossil fuel combustion and deforestation. Scientists predict this will cause global average temperatures to rise. Which of the following best predicts a consequence of rising temperatures on a freshwater lake ecosystem?

A. Warmer water will hold more dissolved oxygen, benefiting fish populations.
B. Increased evaporation will raise water levels, expanding habitat for aquatic species.
C. Warmer temperatures will increase algae growth, potentially reducing oxygen levels and harming fish.
D. Rising temperatures will decrease decomposition rates, allowing more nutrients to accumulate.

Question 53 RC3 — Organisms, Populations, and Ecosystems

Clownfish live among the stinging tentacles of sea anemones. The clownfish are protected from predators by the anemone, while the anemone benefits from the clownfish removing parasites and providing nutrients through their waste. Which ecological relationship does this describe?

A. Parasitism
B. Mutualism
C. Commensalism
D. Predation

Question 54 RC3 — Organisms, Populations, and Ecosystems

Two species of warblers live in the same forest and eat the same type of insects at the same time of day. Which of the following is the most likely long-term outcome for these two species?

A. Both species will develop a mutualistic relationship and share food equally.
B. The two species will merge into a single species over time through interbreeding.
C. One species will outcompete the other or both will evolve to use different resources.
D. The populations will grow together at the same rate because they share the same niche.

Question 55 RC3 — Organisms, Populations, and Ecosystems

After a volcanic eruption covers an area with lava, bare rock is left behind. Over many decades, lichens colonize the rock, followed by mosses, grasses, shrubs, and eventually a forest. This process is best described as which of the following?

A. Primary succession beginning on a lifeless substrate
B. Secondary succession following a major disturbance
C. Ecological recovery driven by the recolonization of surviving species
D. Climax community formation caused by increased rainfall

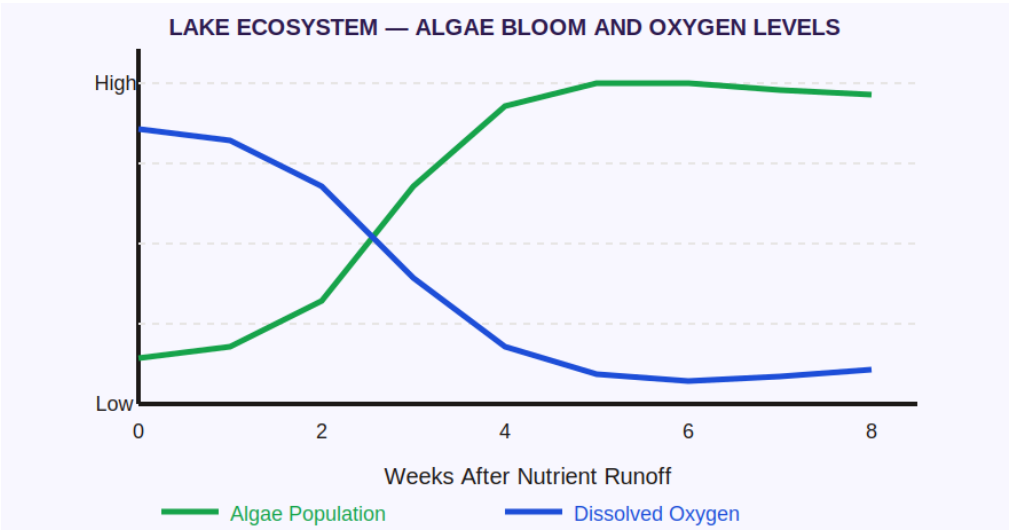
Question 56 RC3 — Organisms, Populations, and Ecosystems

A coral reef ecosystem experiences a prolonged increase in ocean water temperature. Within months, the corals expel the symbiotic algae living in their tissues, turning white — a process called bleaching. Fish populations that depend on the reef begin to decline. Which of the following best explains the chain of events leading to the fish population decline?

A. Bleaching weakens coral structure, destroying fish habitat and reducing food availability.
B. Rising temperatures directly kill the fish by exceeding their thermal tolerance.
C. The loss of algae reduces photosynthesis in the reef, decreasing oxygen levels and suffocating fish.
D. Warmer water increases fish metabolism, requiring them to consume more food than is available.

Question 57 RC3 — Organisms, Populations, and Ecosystems

A lake that normally supports a stable fish population experiences an influx of nutrient runoff from nearby farms. Algae populations explode, blocking sunlight and depleting oxygen as the algae die and decompose. Which of the following best describes the primary limiting factor that causes fish populations to decline in this scenario?



A. Increased predation by algae-eating fish reduces prey availability for larger fish.
B. Oxygen depletion caused by algae decomposition creates conditions that fish cannot survive.
C. The excess nutrients directly poison the fish by entering their tissues.
D. Sunlight reduction caused by algae decreases the water temperature below fish tolerance.

Question 58 RC3 — Organisms, Populations, and Ecosystems

In a savanna ecosystem, the dry season reduces grass cover significantly. During this period, zebra populations decline, followed shortly after by lion populations. When the rainy season returns and grass recovers, zebra populations rebound, followed by lion populations. Which of the following best explains the delayed recovery of lion populations compared to zebra populations?

A. Lions require more water than zebras and must wait for rivers to refill before returning.
B. Lions migrate to different ecosystems during the dry season and take longer to return.
C. Lion reproduction rates are slower than zebra reproduction rates regardless of resource availability.
D. Lion populations recover more slowly because they depend on prey populations that must first increase.

Question 59 RC3 — Organisms, Populations, and Ecosystems

A scientist studying population changes in a wetland ecosystem collects data over 10 years showing that a certain frog species has declined by 60%. She concludes that habitat loss is the primary cause. Which of the following would most strengthen her conclusion?

A. Observing that the frog species is also declining in a different country.
B. Collecting data showing that frog populations declined before habitat loss began.
C. Interviewing local residents about changes they have observed in the wetland.
D. Finding that frog populations in undisturbed wetlands nearby have remained stable.

Question 60 RC3 — Organisms, Populations, and Ecosystems

A researcher claims that introducing a non-native plant species to a nature preserve will increase overall biodiversity. A second researcher argues that non-native species typically reduce native biodiversity by outcompeting native plants. Which of the following would be the most scientifically valid way to evaluate these competing claims?

A. Accept the first researcher's claim because introducing species can only add diversity, not reduce it.
B. Survey the scientific literature to find studies supporting whichever researcher has more publications.
C. Design a controlled study comparing biodiversity in areas with and without the non-native plant species.
D. Allow the non-native plant to be introduced and observe results without a control group.

ANSWER SHEET — FOR TEACHER USE ONLY

Student Name: _____ ID: _____ Teacher: _____

Q	A	B	C	D	Q	A	B	C	D	Q	A	B	C	D	Q	A	B	C	D
1	O	O	O	O	16	O	O	O	O	31	O	O	O	O	46	O	O	O	O
2	O	O	O	O	17	O	O	O	O	32	O	O	O	O	47	O	O	O	O
3	O	O	O	O	18	O	O	O	O	33	O	O	O	O	48	O	O	O	O
4	O	O	O	O	19	O	O	O	O	34	O	O	O	O	49	O	O	O	O
5	O	O	O	O	20	O	O	O	O	35	O	O	O	O	50	O	O	O	O
6	O	O	O	O	21	O	O	O	O	36	O	O	O	O	51	O	O	O	O
7	O	O	O	O	22	O	O	O	O	37	O	O	O	O	52	O	O	O	O
8	O	O	O	O	23	O	O	O	O	38	O	O	O	O	53	O	O	O	O
9	O	O	O	O	24	O	O	O	O	39	O	O	O	O	54	O	O	O	O
10	O	O	O	O	25	O	O	O	O	40	O	O	O	O	55	O	O	O	O
11	O	O	O	O	26	O	O	O	O	41	O	O	O	O	56	O	O	O	O
12	O	O	O	O	27	O	O	O	O	42	O	O	O	O	57	O	O	O	O
13	O	O	O	O	28	O	O	O	O	43	O	O	O	O	58	O	O	O	O
14	O	O	O	O	29	O	O	O	O	44	O	O	O	O	59	O	O	O	O
15	O	O	O	O	30	O	O	O	O	45	O	O	O	O	60	O	O	O	O

SCORING SUMMARY

	Correct	Total	Percent
RC1 — Molecular & Cellular Biology	_____	21	_____%
RC2 — Classification, Heredity & Evolution	_____	15	_____%
RC3 — Organisms, Populations & Ecosystems	_____	24	_____%
TOTAL	_____	**60**	**_____%**